Alcohols & Phenols,

By

Prof.(DR.) RUPESH KUMAR

Department of Chemistry , Kirori Mal College

University of Delhi, Delhi -110007

जननी जन्मभूमि, स्वर्ग से महान है।
इसके वास्ते ये तन है, मन है और प्राण हैं।।

इसके कण-कण में लिखा राम-कृष्ण नाम है।
हुतात्माओं के रूधिर से भूमि शस्य-श्याम है।
धर्म का ये धाम है, सदा इसे प्रणाम है।
स्वतंत्र है धरा यहां, स्वतंत्र आसमान है।
जननी जन्मभूमि, स्वर्ग से महान है।।1।।

इसकी आन पर कभी जो बात कोई आ पड़े।
इसके सामने जो जुल्म के पहाड़ हों खड़े।।
शत्रु सब जहान हों, विरूद्ध विधि-विधान हो,
मुकाबला करेंगे जब तक जान में ये जान है।
जननी जन्मभूमि, स्वर्ग से महान है।।2।।

इसकी गोद में हजारों गंगा-यमुना झूमती।
इसके पर्वतों की चोटियां गगन को चूमती।।
भूमि ये महान है, निराली इसकी शान है।
इसकी जय-पताका ही स्वयं विजय-निशान है।
जननी जन्मभूमि, स्वर्ग से महान है।।3।।

Unit - IV: Alcohols, Phenols, Ethers & Epoxides (10 Hours)

Alcohols: Relative reactivity of 1°, 2°, 3° alcohols, reactions of alcohols with sodium,HX (Lucas test), esterification, oxidation (with PCC, alkaline KMnO4, acidic dichromate,conc. HNO3). Oppenauer oxidation; Diols: oxidation of diols by periodic acid and lead tetraacetate, Pinacol-Pinacolone rearrangement.

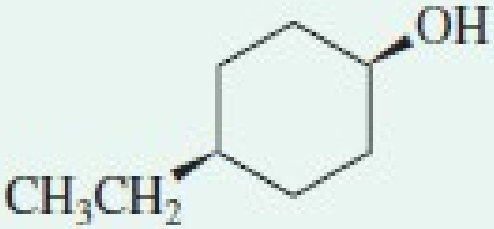
Phenols: Preparation using Cumene hydroperoxide, Acidity and factors affecting it,
Kolbe's–Schmidt reactions, Riemer-Tiemann reaction, Houben–Hoesch condensation,
Schotten–Baumann reaction, Fries and Claisen rearrangements and their mechanism.

Ethers and Epoxides: Acid and Base catalyzed cleavage reactions.

Draw the structures of the following alcohols: (a) (S)-3-methyl-3-hexanol; (b) trans-2-bromocyclopentanol; (c) 2,2-dimethyl-1-propanol (neopentyl alcohol).

Name the following compounds.

(a) $CH_3CH(CH_3)CH_2CH(OH)CH_3$

(b) 

(c) $CH_3CH(Br)CH(Cl)CH_2OH$

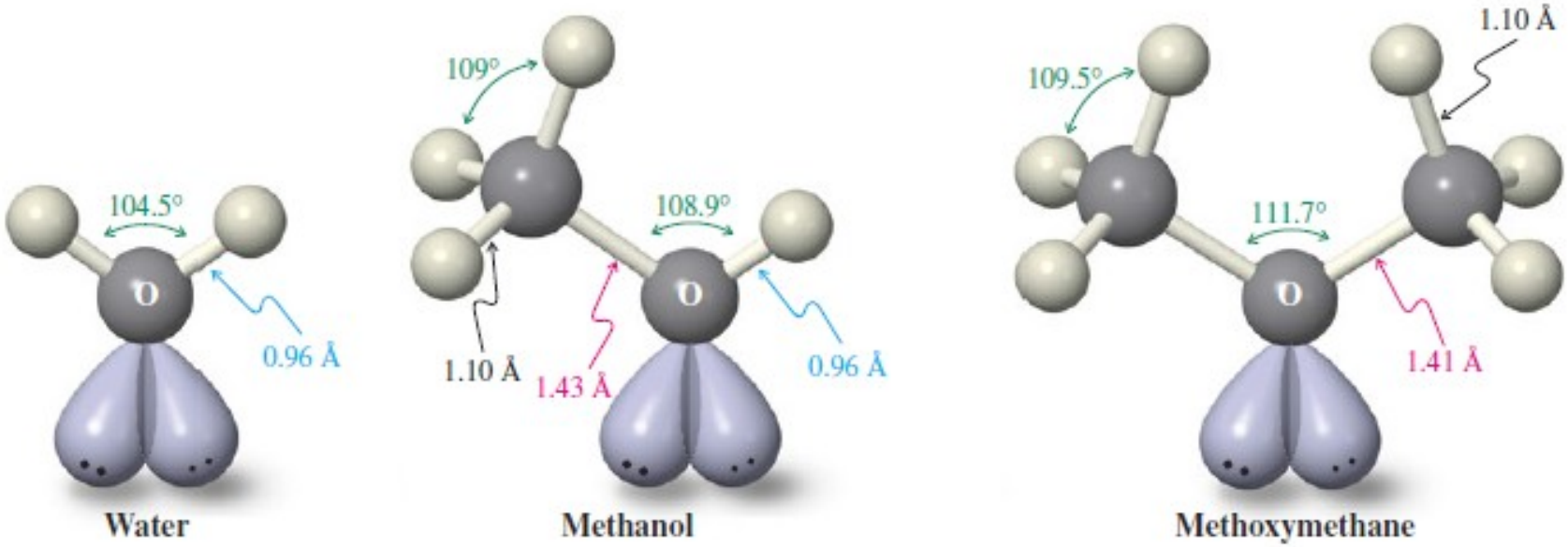


Figure 8-1 The similarity in structure of water, methanol, and methoxymethane.

The consequences of polarization are seen clearly in the electrostatic potential maps of water and methanol.

Bond and Molecular Dipoles of Water and Methanol

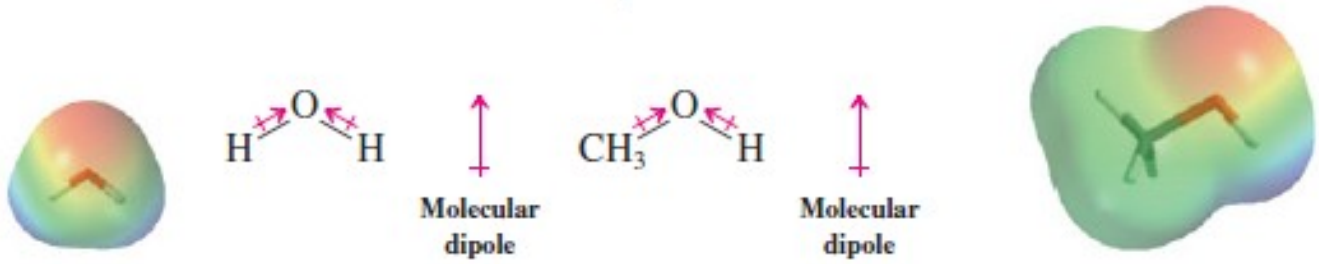


Table 8-1 Physical Properties of Alcohols and Selected Analogous Haloalkanes and Alkanes

Compound	IUPAC name	Common name	Melting point (°C)	Boiling point (°C)	Solubility in H_2O at 23°C
CH_3OH	Methanol	Methyl alcohol	−97.8	65.0	Infinite
CH_3Cl	Chloromethane	Methyl chloride	−97.7	−24.2	0.74 g/100 mL
CH_4	Methane		−182.5	−161.7	3.5 mL (gas)/100 mL
CH_3CH_2OH	Ethanol	Ethyl alcohol	−114.7	78.5	Infinite
CH_3CH_2Cl	Chloroethane	Ethyl chloride	−136.4	12.3	0.447 g/100 mL
CH_3CH_3	Ethane		−183.3	−88.6	4.7 mL (gas)/100 mL
$CH_3CH_2CH_2OH$	1-Propanol	Propyl alcohol	−126.5	97.4	Infinite
$CH_3CH_2CH_3$	Propane		−187.7	−42.1	6.5 mL (gas)/100 mL
$CH_3CH_2CH_2CH_2OH$	1-Butanol	Butyl alcohol	−89.5	117.3	8.0 g/100 mL
$CH_3(CH_2)_4OH$	1-Pentanol	Pentyl alcohol	−79	138	2.2 g/100 mL

- **The effect is even more pronounced in water, which has two hydrogens available for hydrogen bonding (see Figure 8-2).**
- **This phenomenon explains why water, with a molecular weight of only 18, has a boiling point of 100 C.**
- **Without this property, water would be a gas at ordinary temperatures. Considering the importance of water in all living organisms, imagine how the absence of liquid water would have affected the development of life on our planet.**

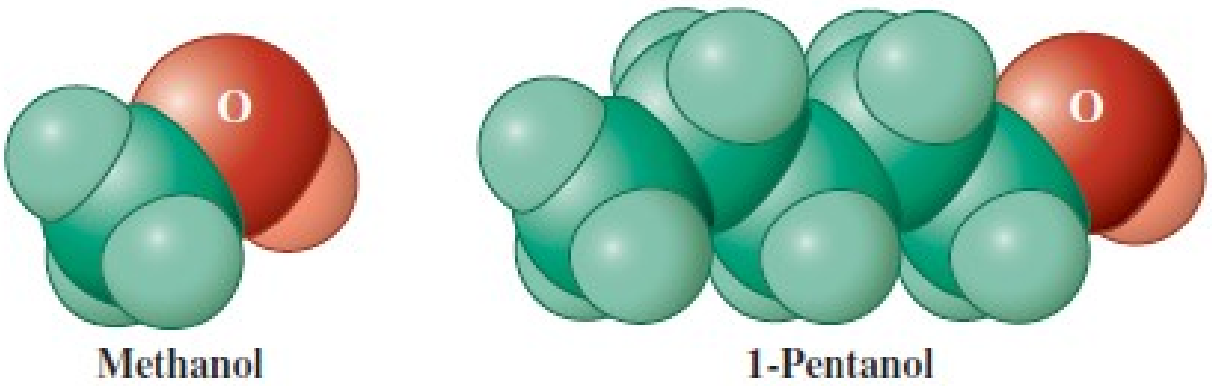


Figure 8-3 The hydrophobic and hydrophilic parts of methanol and 1-pentanol (space-filling models). The polar functional group dominates the physical properties of methanol: The molecule is completely soluble in water but only partially so in hexane. Conversely, the increased size of the hydrophobic part in 1-pentanol leads to infinite solubility in hexane but reduced solubility in water (Table 8-1).

Table 8-2 lists the pKa values of several alcohols. A comparison of these values with those given in Table 2-2 for mineral and other strong acids shows that alcohols, like water, are fairlyweak acids. Their acidity is far greater, however, than that of alkanes and haloalkanes.

Table 8-2 pK_a Values of Alcohols in Water

Compound	pK_a	Compound	pK_a
HOH	15.7	$ClCH_2CH_2OH$	14.3
CH_3OH	15.5	CF_3CH_2OH	12.4
CH_3CH_2OH	15.9	$CF_3CH_2CH_2OH$	14.6
$(CH_3)_2CHOH$	17.1	$CF_3CH_2CH_2CH_2OH$	15.4
$(CH_3)_3COH$	18		

It is sometimes suffi cient to generate alkoxides in less than stoichiometric equilibrium concentrations. For this purpose, we may add an alkali metal hydroxide to the alcohol.

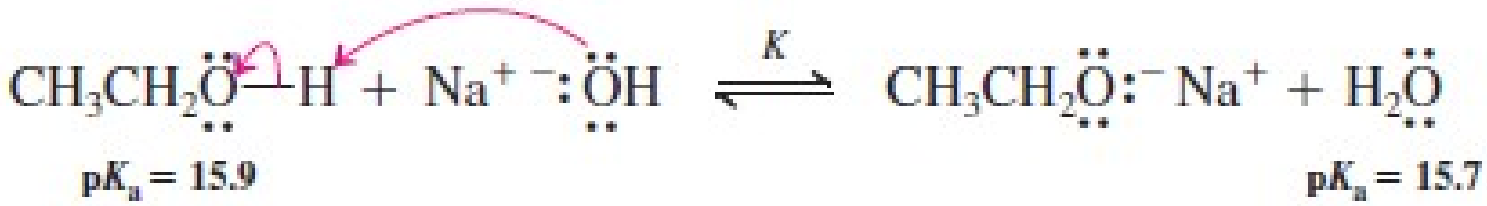
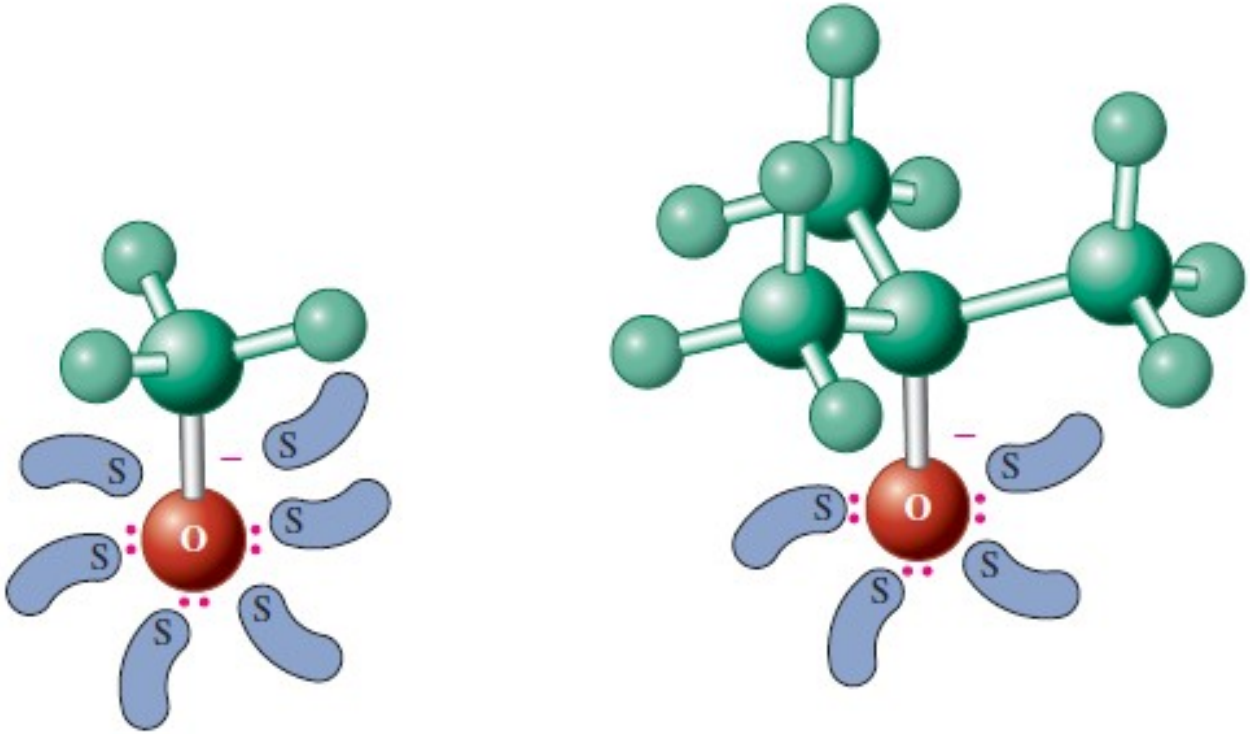
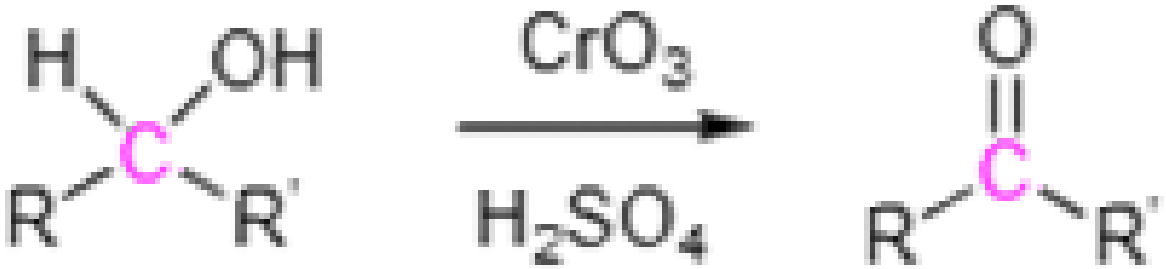
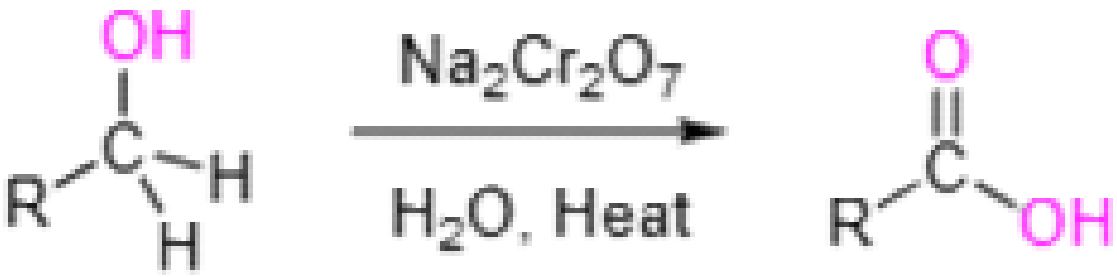


Figure 8-4 The smaller methoxide ion is better solvated than is the larger tertiary butoxide ion. (S = solvent molecules).

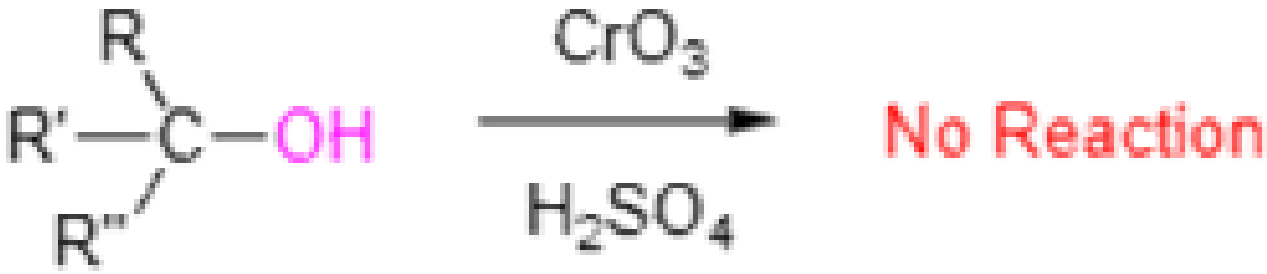




2° Alcohol **Ketone**

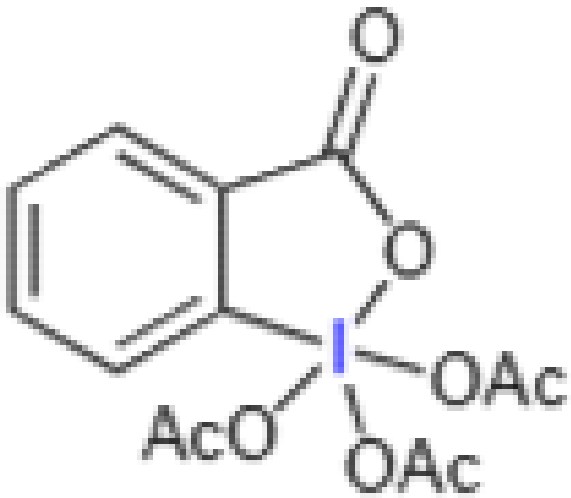


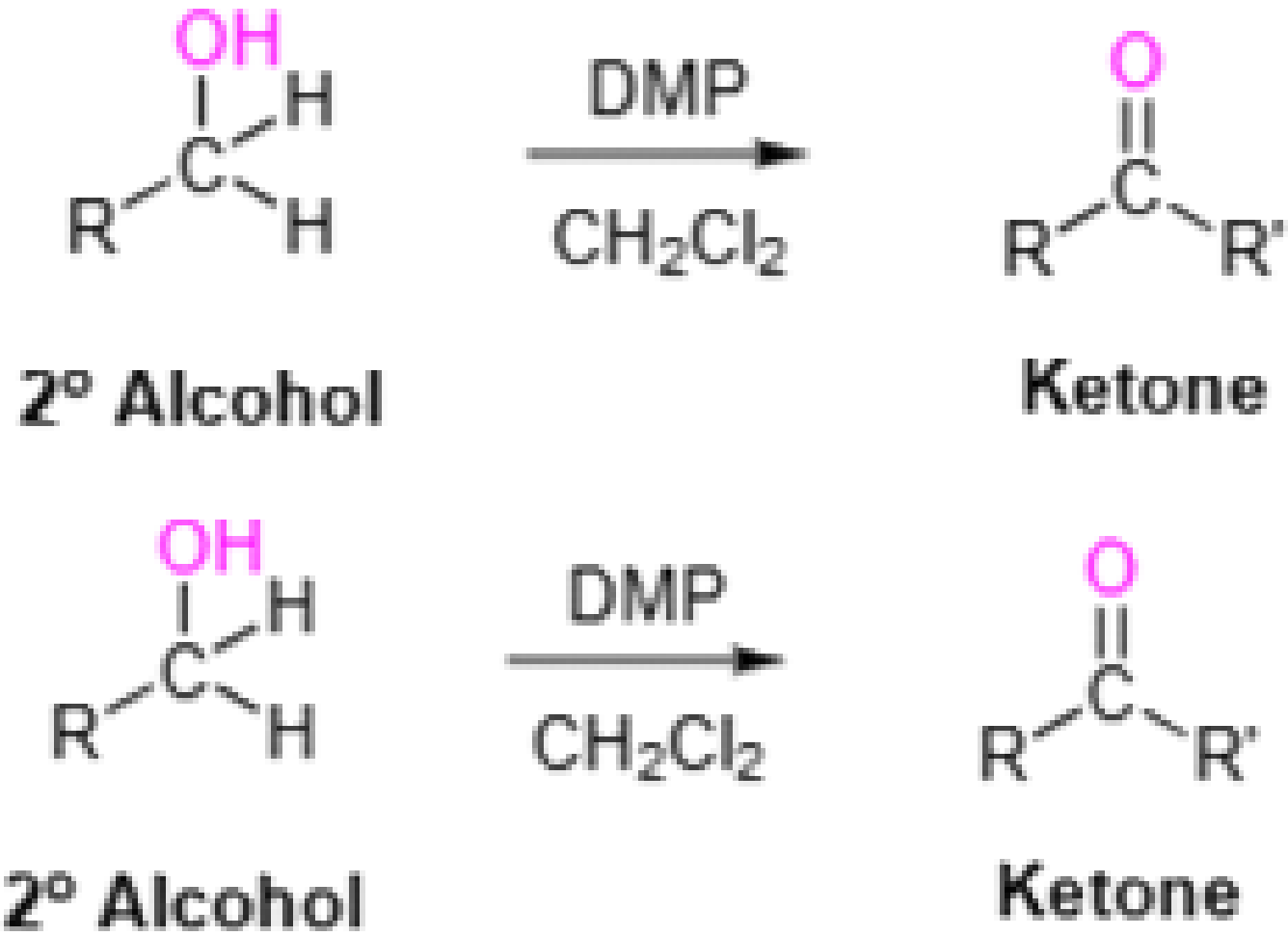
1° Alcohol **Carboylic Acid**



3° Alcohol

PCC is being replaced in laboratories by Dess- Martin periodinane (DMP) in dichloromethane solvent, which has several practical advantages over PCC, such as producing higher yields and requiring less rigorous conditions (lower reaction temperature and a nonacidic medium). DMP is named after Daniel Dess and James Martin, who developed it in 1983.





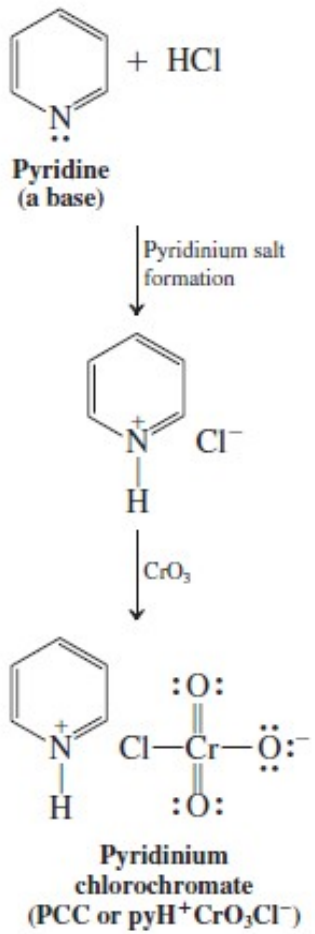
Testing breath for the presence of alcohol.

$$\underset{\text{Orange}}{2\,K_2Cr_2O_7} + 8\,H_2SO_4 + 3\,CH_3CH_2OH \longrightarrow$$

$$\underset{\text{Green}}{2\,Cr_2(SO_4)_3} + 2\,K_2SO_4 + 3\,CH_3\overset{\overset{\displaystyle O}{\|}}{C}OH + 11\,H_2O$$

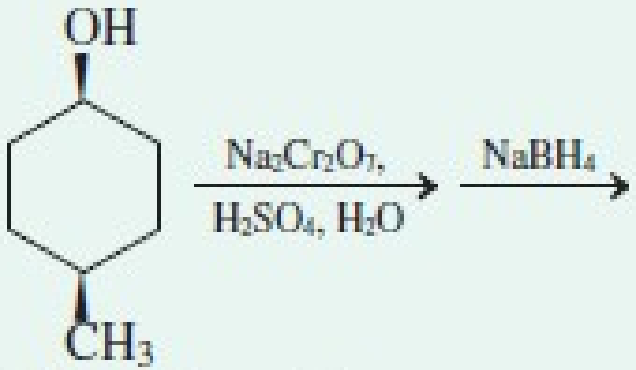
- **If green develops beyond the halfway mark, a blood alcohol concentration greater than 0.08% is indicated, which is considered a criminal offense in many countries.**
- **A more sophisticated version of this simple procedure uses spectrophotometric techniques to quantify the extent of oxidation, the so-called breath analyzer test, and more recent developments include the use of mini – gas chromatographs, electrochemical analyzers, and infrared spectrometers.**

a water-free form of Cr(VI) has been developed by reaction of CrO_3 with HCl, followedby the addition of the organic base pyridine. The result is the oxidizing agent pyridinium chlorochromate, abbreviated as py CrO_3Cl_2 or just PCC (margin), which gives excellent yields of aldehydes upon exposure to primary alcohols in dichloromethane solvent.



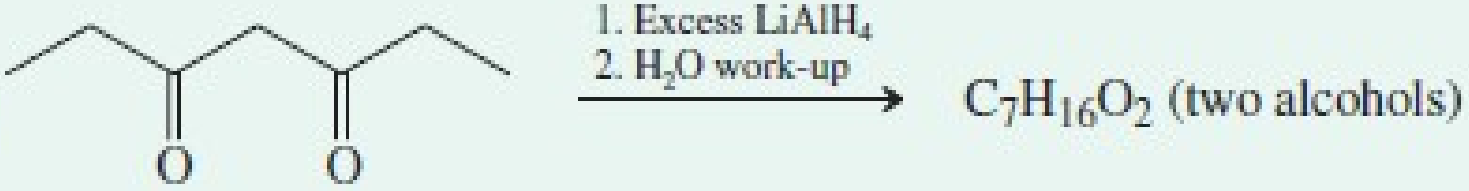
Formulate the product(s) of each of the following steps. What can you say about stereochemistry?

(a)

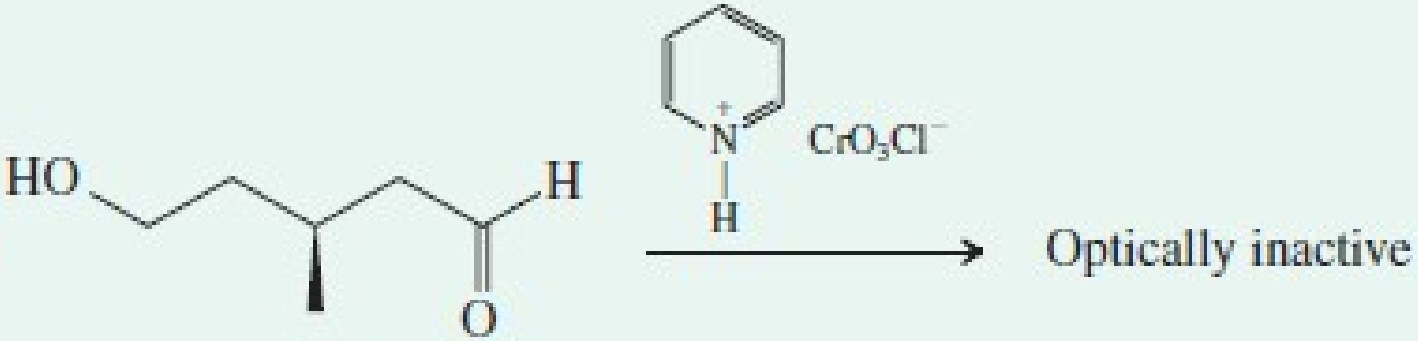


***cis*-4-Methylcyclohexanol**

(b)



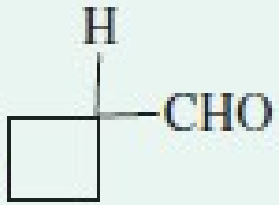
(c)

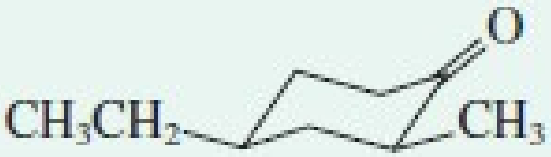


Optically active

Formulate a synthesis of each of the following carbonyl compounds from the corresponding alcohol.

(a) $CH_3CH_2\overset{\overset{\displaystyle O}{\|}}{C}CH(CH_3)_2$

(b) 

(c) 

Oppenauer Oxidation

Mechanism

1. In the first step, alcohol coordinates with aluminium isopropoxide to form a complex

2. This complex reacts with ketone to form a six-membered transition complex

3. The alpha-carbon of the alcohol is converted to the carbonyl carbon from the aluminiumcatalyzed hydride shift.

4. The acetone proceeds over a six-membered transition state.

5. The desired ketone is formed after the hydride transfer

This reaction is used to oxidize alcohols to carbonyl compounds. For example, simple ketone acetone or cyclohexanones is used as the hydride acceptor in the presence of aluminium alkoxide usually isopropoxide or t-butoxide. The oxidation is the exact reverse of Meerwein-Ponndorf-Verley reduction and involves deprotonation of the alcohol by equilibration with the alkoxide followed by hydride transfer. This equilibrium is generally displaced to the right by using a large excess of the hydride acceptor.

Alkoxide Formation with Methyllithium

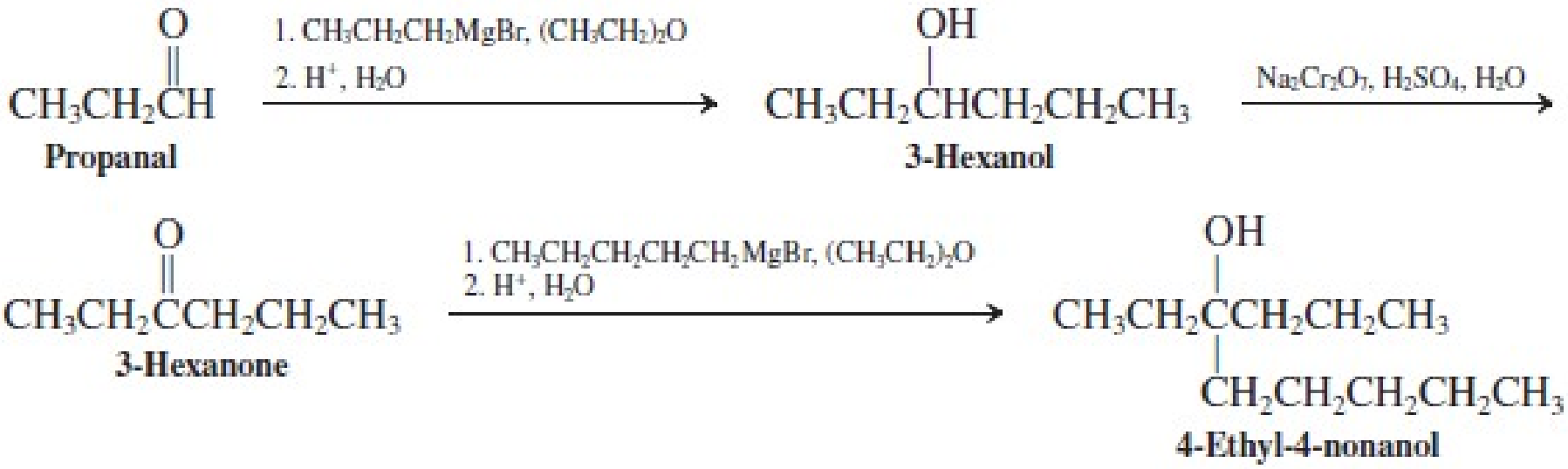
1,1-Dimethylethanol (*tert*-Butylalcohol) $pK_a = 18$

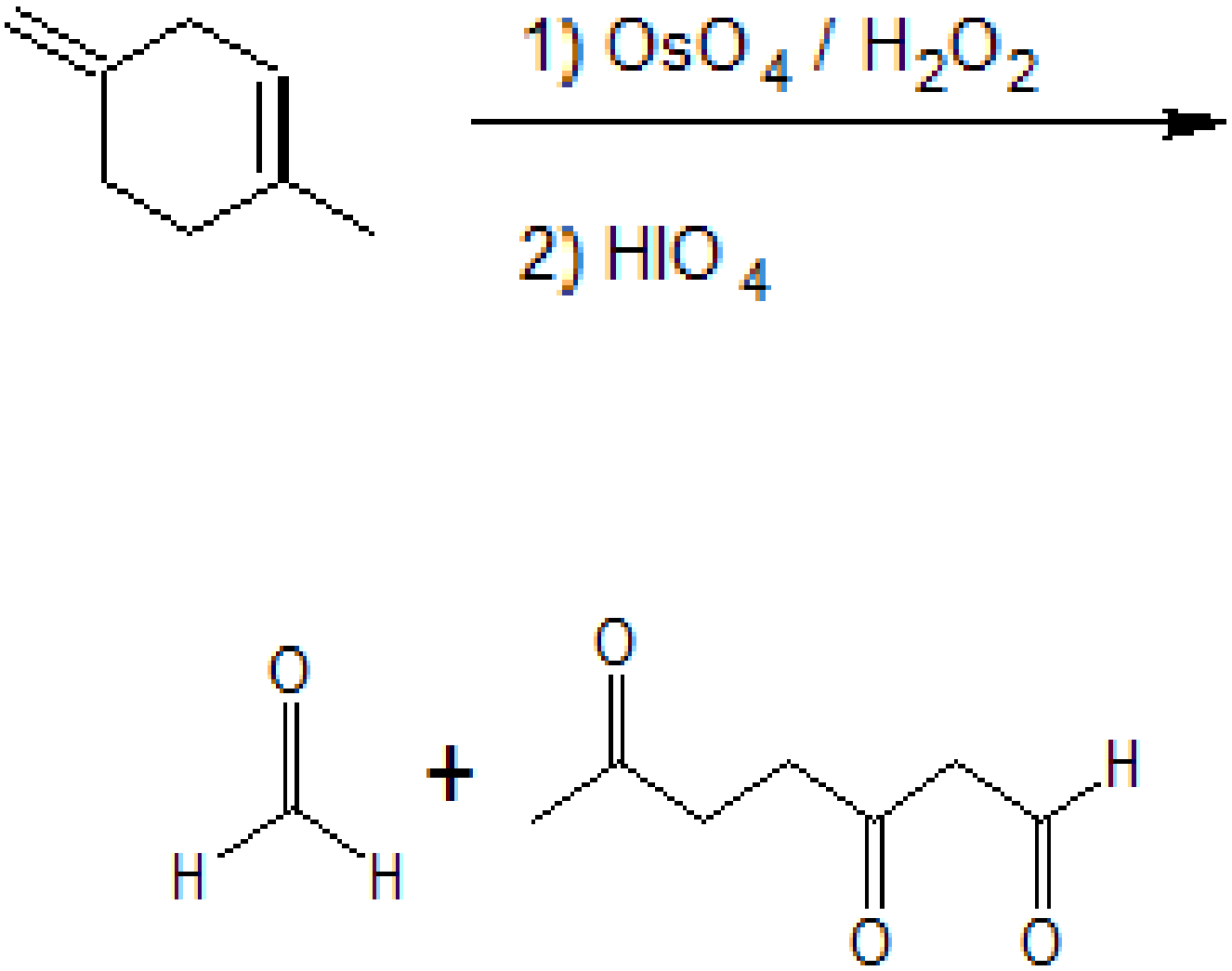
Methyllithium

Lithium *tert*-butoxide

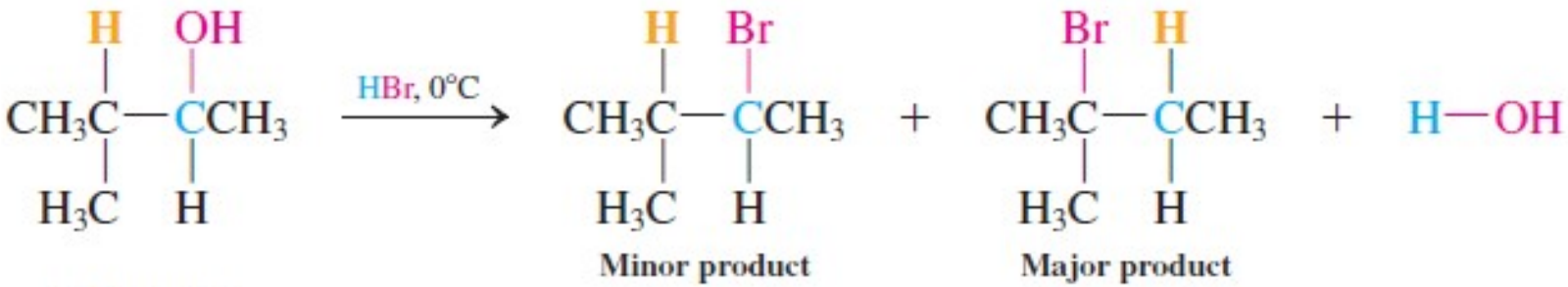
Methane $pK_a \simeq 50$

Synthesis of 4-Ethyl-4-nonanol



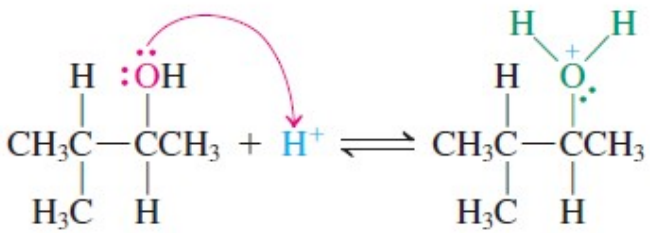

1) OsO 4 / H2O2
2) HIO 4
O
H
H
+
O
H
O
O

Hydride Shift in the S_N1 Reaction of an Alcohol with HBr

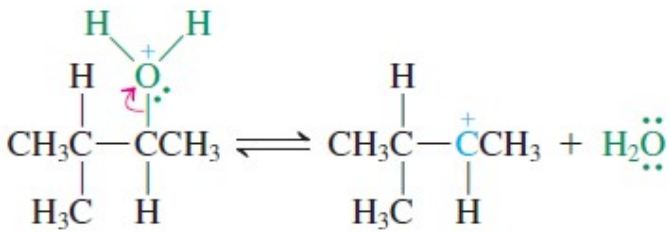


Mechanism of Carbocation Rearrangement

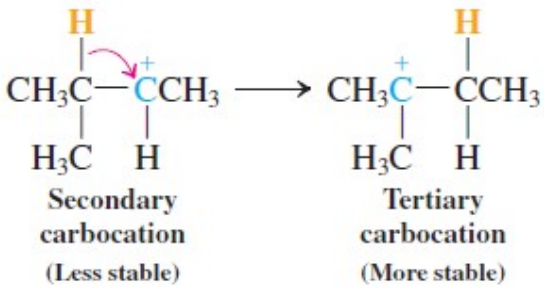
Step 1. Protonation



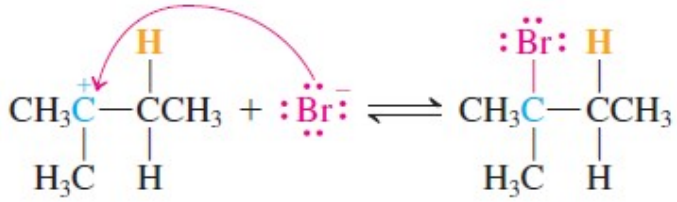
Step 2. Loss of water



Step 3. Hydride shift

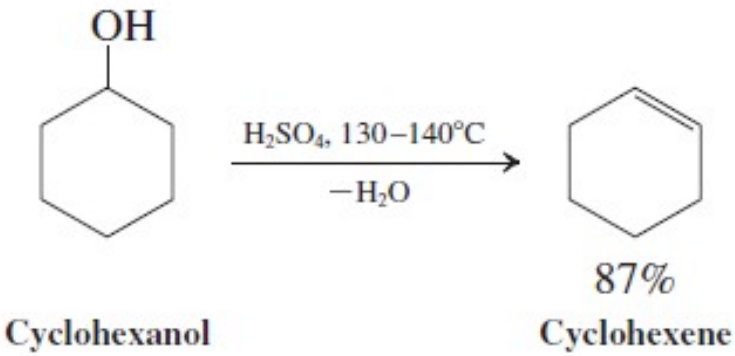


Step 4. Trapping by bromide

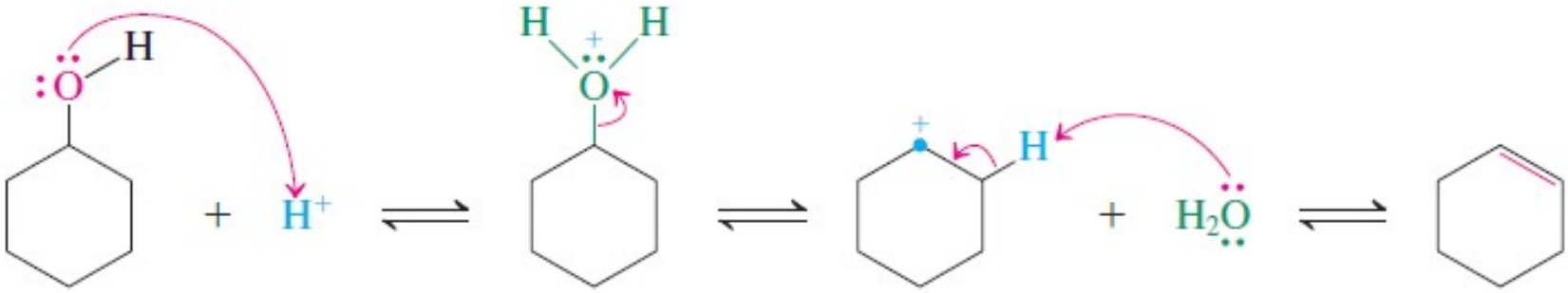


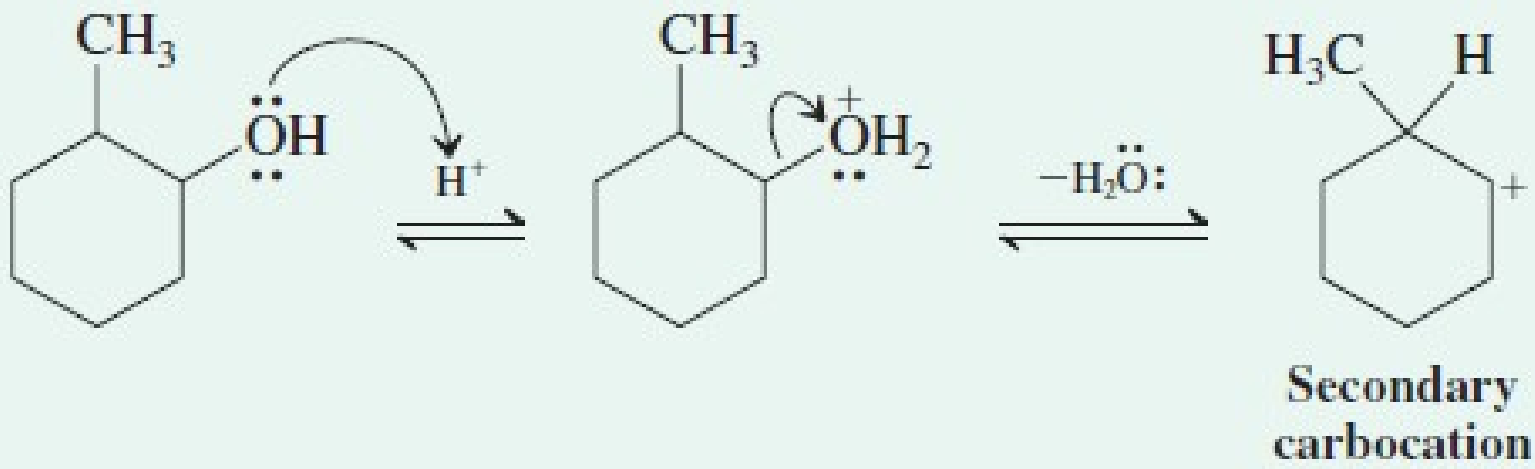
The Lucas test is a, now often obsolete, chemical method using Lucas reagent (anhydrous \($ZnCl_2$) in concentrated (HCl) to classify low-molecular-weight alcohols as primary, secondary, or tertiary. It works by measuring the rate of turbidity (cloudiness) caused by the formation of insoluble alkyl chlorides, with tertiary alcohols reacting instantly, secondary within 5–10 minutes, and primary not at all at room

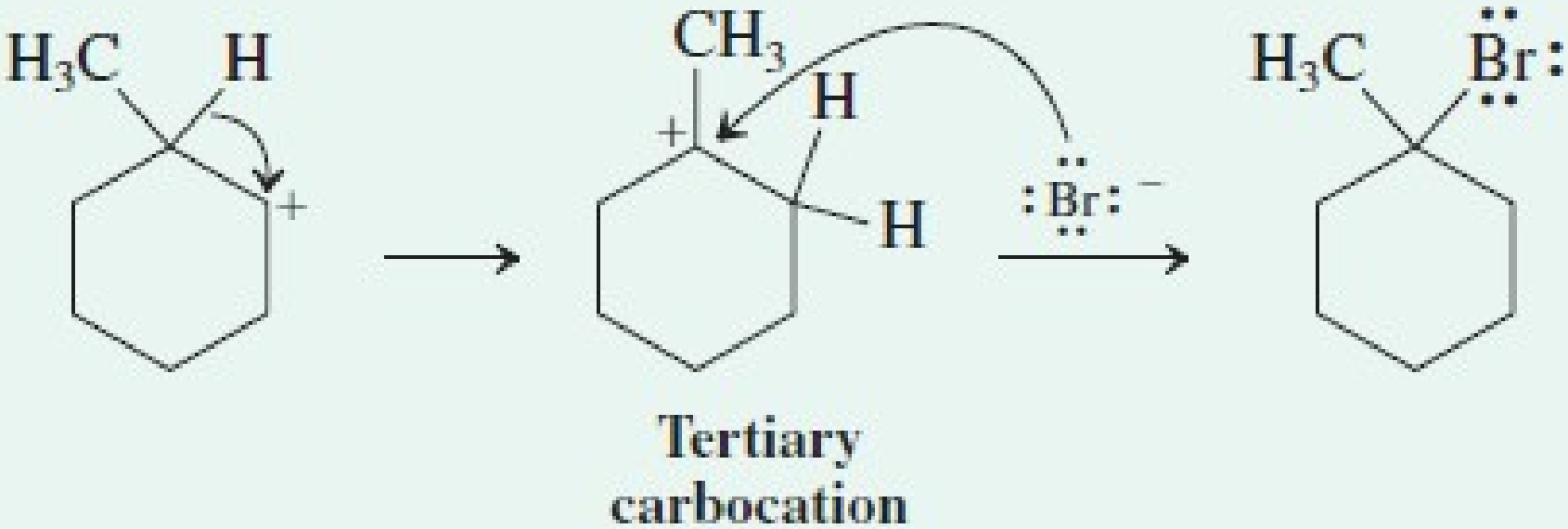
Alcohol Dehydration by the E1 Mechanism



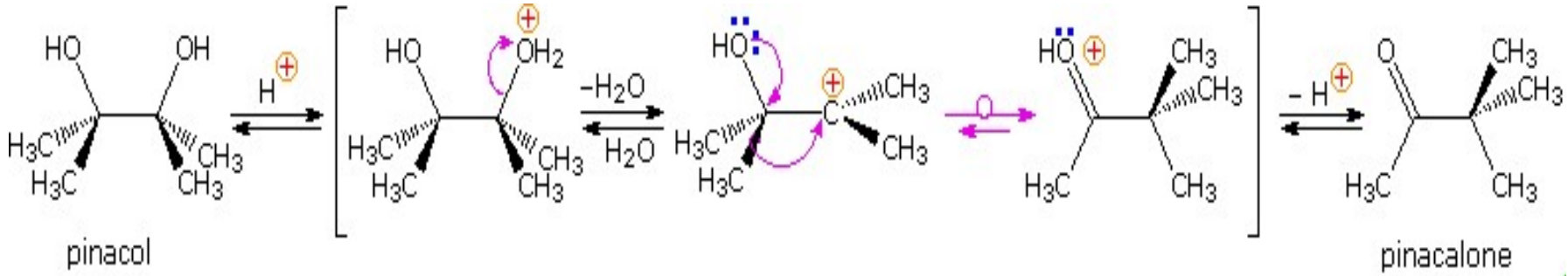
Mechanism of the Dehydration of Cyclohexanol



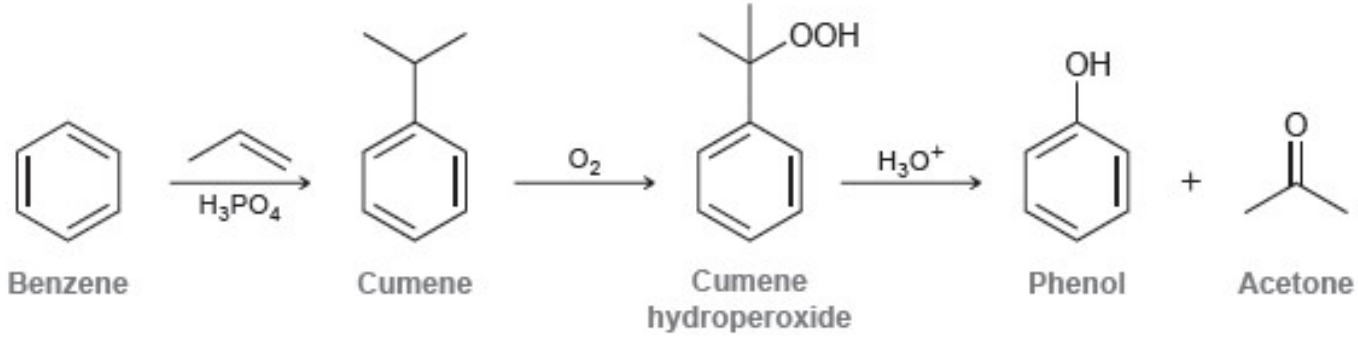

CH3
OH
H+
CH3
OH2
−H2O:
H3C
H
Secondary carbocation


H3C
H
CH3
H
H
:Br:−
H3C
Br:
Tertiary carbocation

- Since the diol is symmetrical, protonation and loss of water takes place with equal probability at either hydroxyl group.
- The resulting 3°-carbocation is relatively stable, and has been shown to return to pinacol by reaction in the presence of isotopically labeled water.
- A 1,2-methyl shift generates an even more stable carbocation in which the charge is delocalized by heteroatom resonance.
- Indeed, this new cation is simply the conjugate acid of the ketone pinacolone, which is the product of repeated rearrangements catalyzed by proton transfer.
- Each step in this rearrangement is potentially reversible, as demonstrated by the acid catalyzed dehydration of pinacolone (and pinacol) to 2,3-dimethyl-1,3-butadiene under vigorous conditions.

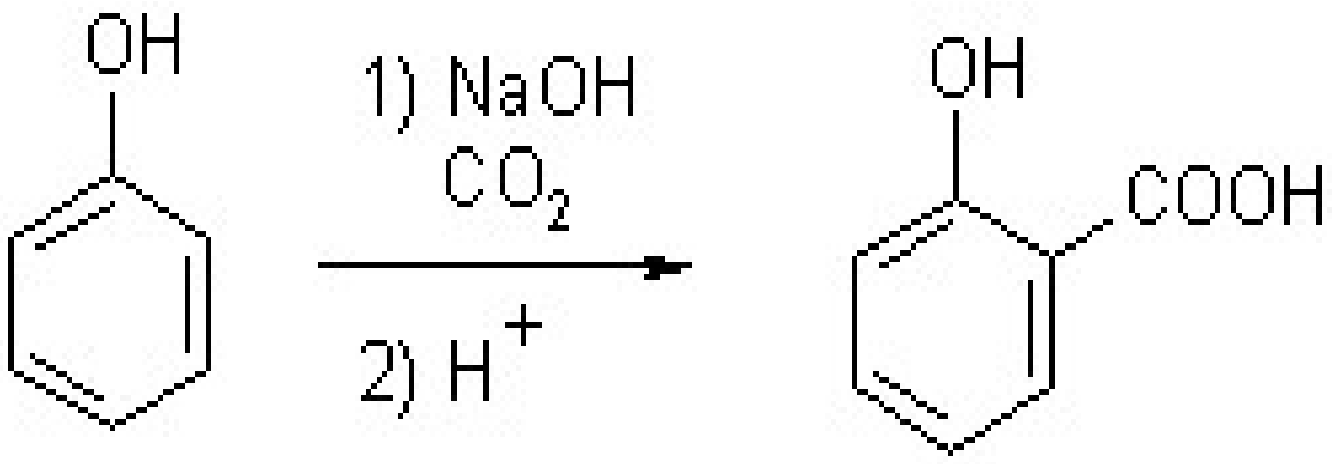


Phenol is prepared industrially via a multistep process involving the formation and oxidation of cumene.



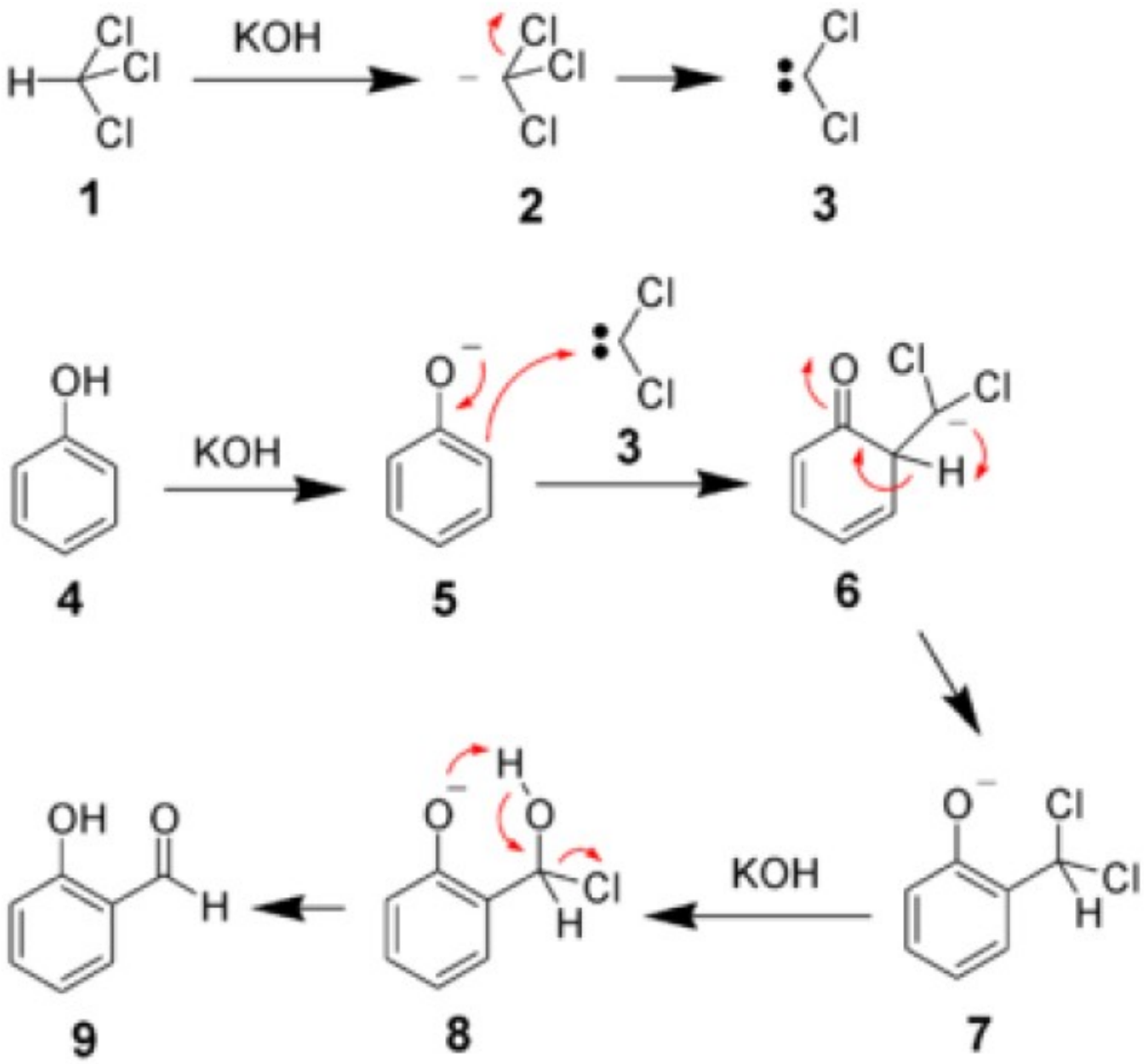
by-product of this process is acetone, which is also a commercially important compound.Over two million tons of phenol is produced each year in the United States. Phenol is used as a precursor in the synthesis of a wide variety of pharmaceuticals and other commercially useful compounds, including bakelite (a synthetic polymer made from phenol and formaldehyde), adhesives for plywood, and antioxidant food additives (BHT and BHA,)

Kolbe-Schmitt Reaction



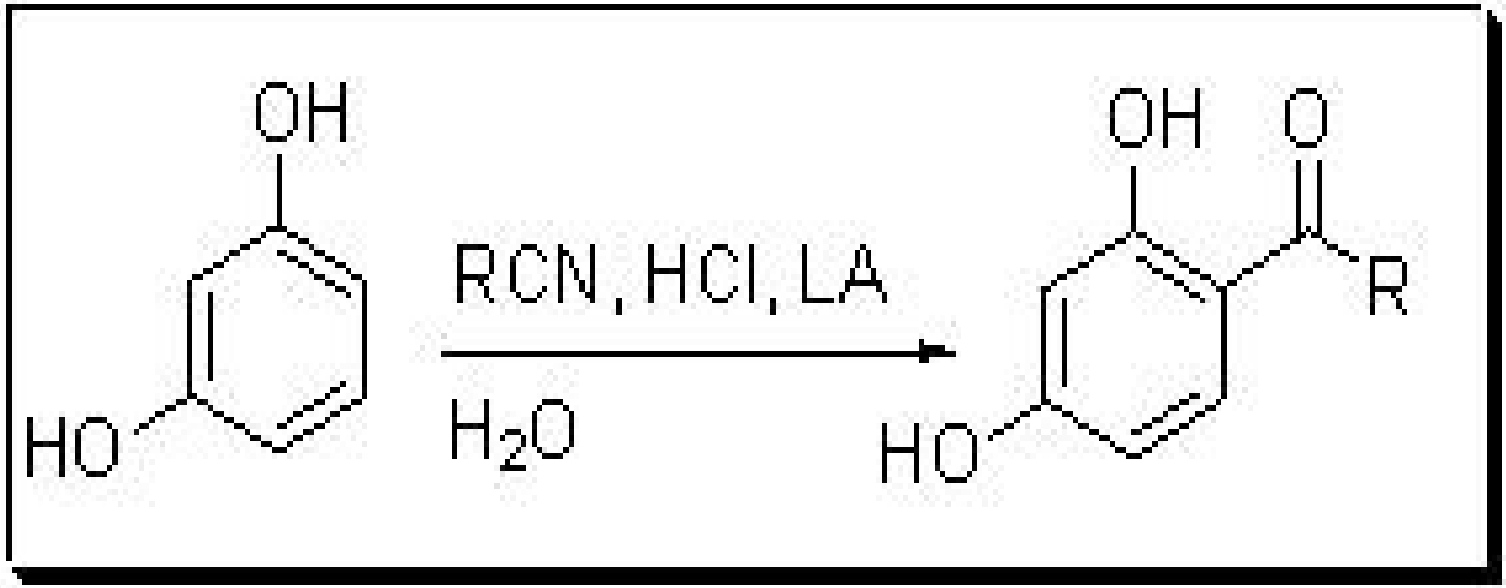
A base-promoted carboxylation of phenols that allows the synthesis of salicylic acid derivatives.

The Reimer-Tiemann reaction is an organic reaction that converts phenols into ortho-hydroxybenzaldehydes (mainly salicylaldehyde) using chloroform ($CHCl_3$) and a strong base (NaOH or KOH), followed by acid hydrolysis. It is a form of electrophilic aromatic substitution that inserts a formyl group (CHO) onto the aromatic ring



The mechanism of the Reimer-Tiemann reaction

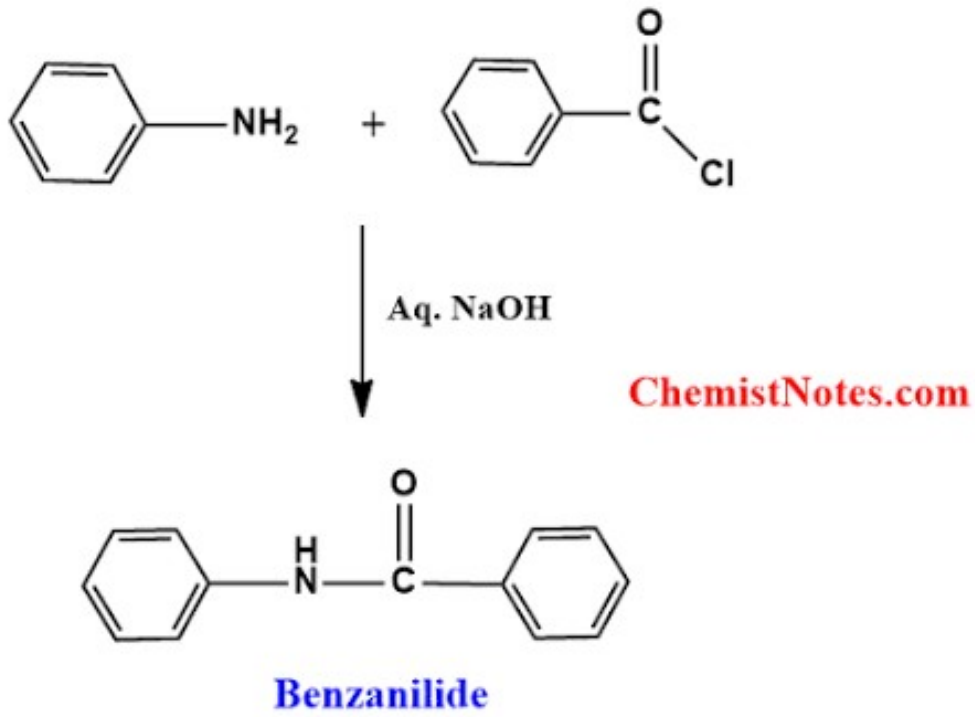
This reaction refers the acylation of phenols using nitrile, hydrochloric acid and Lewis acid.



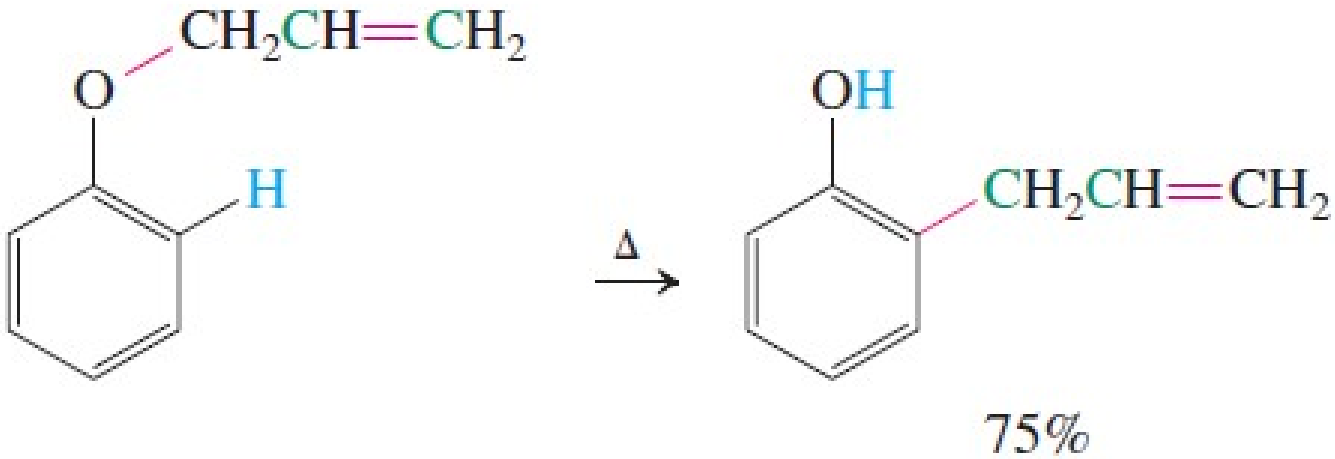
The Houben-Hoesch reaction is the acid-promoted electrophilic substitution between aromatic compounds and nitriles yielding aromatic ketones. It is a variant of the Friedel-Crafts acylation. The reaction was discovered by Kurt Hoesch in 1915, and was developed further by Josef Houben in 1926.

The Schotten-Baumann reaction in phenols is a method for synthesizing phenyl esters by reacting phenol with an acid chloride in the presence of an aqueous base. While the reaction is also widely used for preparing amides from amines, its application to phenols specifically refers to benzoylation.

Schotten-Baumann Reaction Example



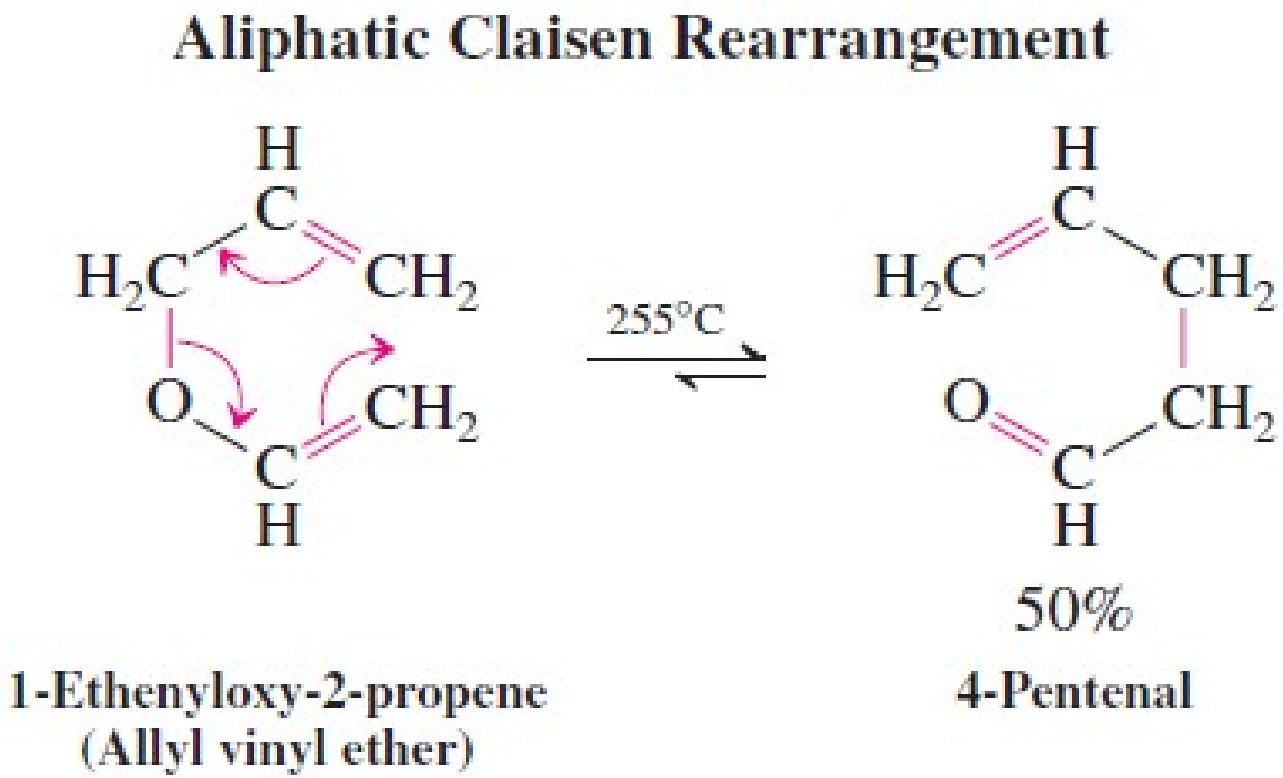
At 200 C, 2-propenyloxybenzene (allyl phenyl ether) undergoes an unusual reaction that leads to the rupture of the allylic ether bond: The starting material rearranges to 2-(2-propenyl)phenol (o-allylphenol).

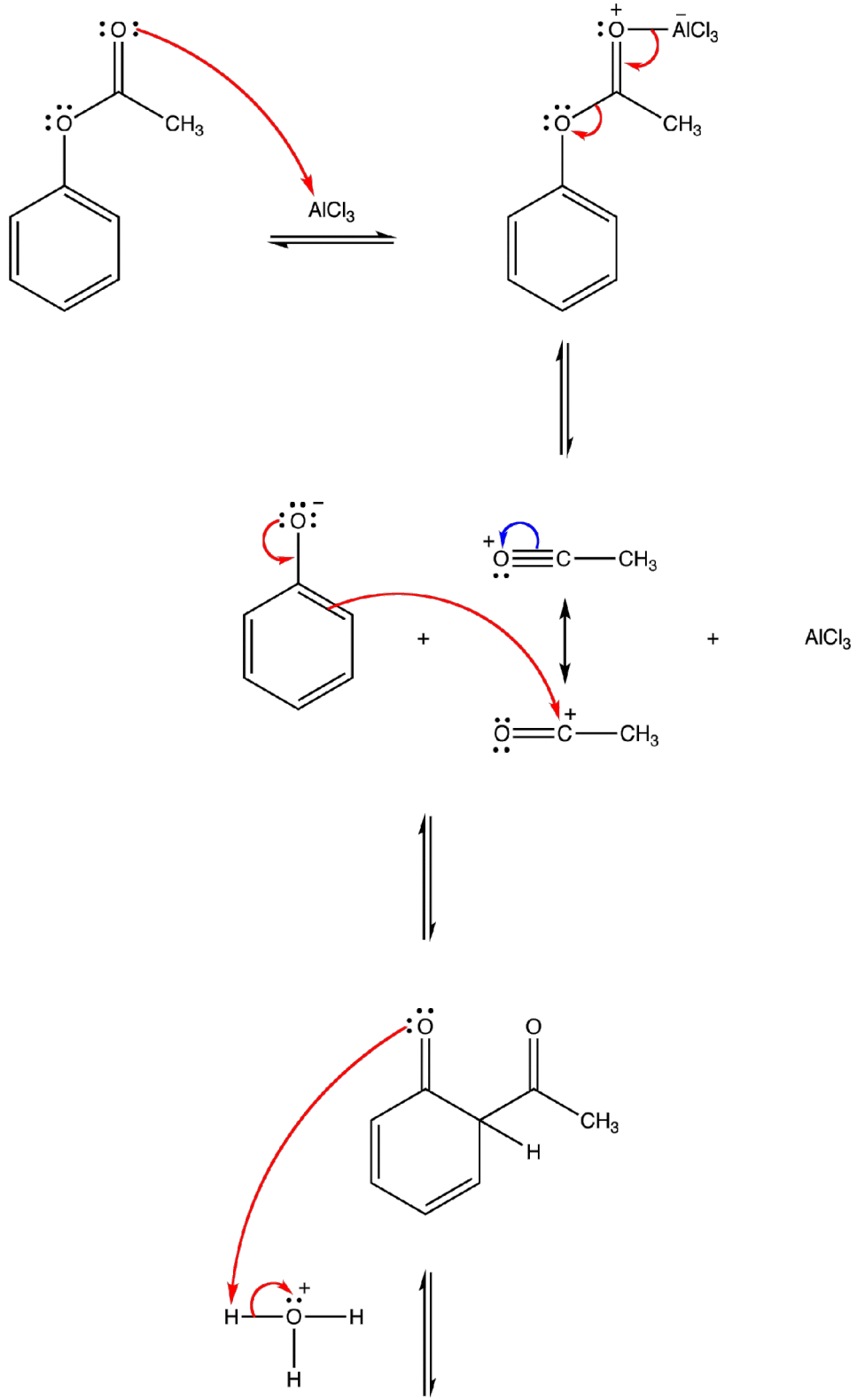


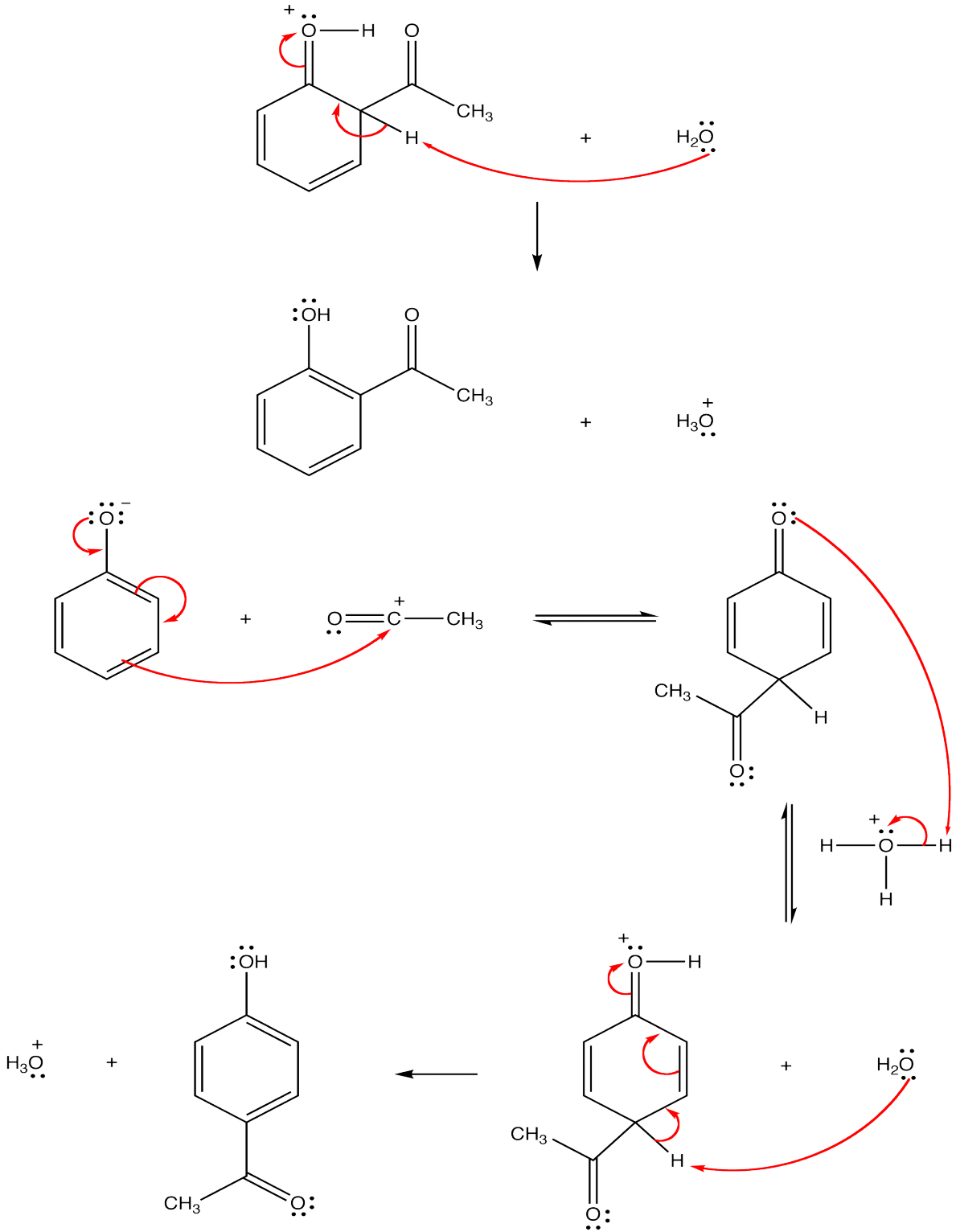
2-Propenyloxybenzene
(Allyl phenyl ether)

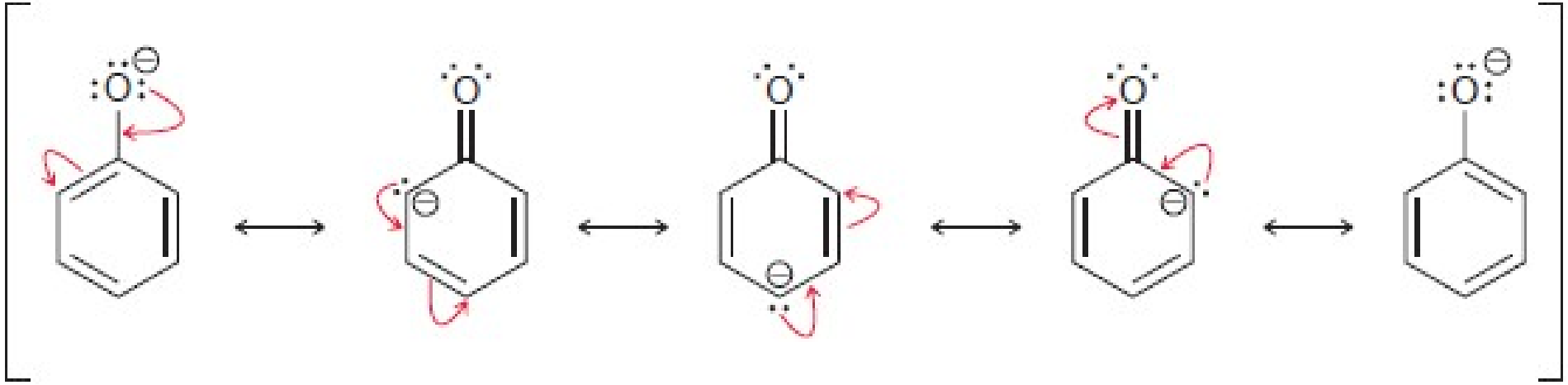
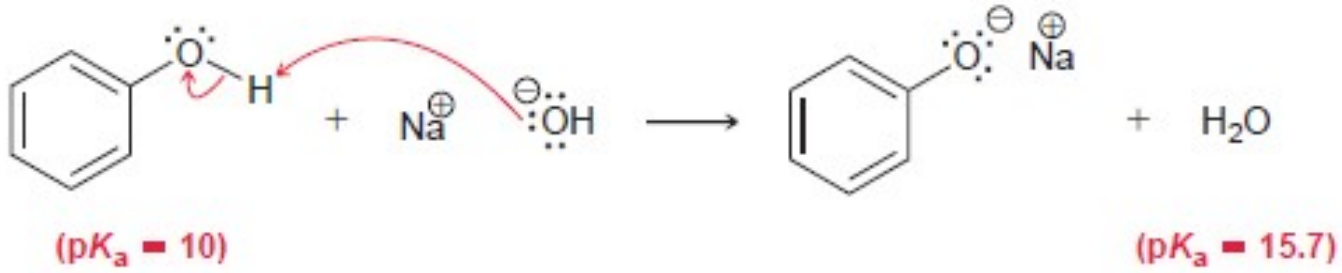
2-(2-Propenyl)phenol
(*o*-Allylphenol)

The Claisen rearrangement is general for other systems. With the nonaromatic 1- ethenyloxy-2-propene (allyl vinyl ether), it stops at the carbonyl stage because there is no driving force for enolization. This is called the aliphatic Claisen rearrangement.




AlCl3
CH3
AlCl3
CH3
CH3
+
AlCl3
CH3
CH3
H
H
H
H


O
H
O
CH3
H
+
H2O
OH
O
CH3
+
H3O
O
+
O
C
CH3
O
CH3
H
O
H
O
H
H
H
O
H
CH3
H
O
+
H2O
H3O
+
OH
CH3
O



This resonance-stabilized anion is called a phenolate, or a phenoxide ion. Resonance stabilization of the phenoxide ion explains why phenol is eight orders of magnitude (100,000,000 times) more acidic than cyclohexanol. As a result, phenol does not need to be deprotonated with a very strong base like sodium hydride. Instead, it can be deprotonated by hydroxide.